Count the little wiggles

How Speakers Turn Waves Into Sound

Name _____ Date _____

Show your thinking with numbers, marks, or words.

1. Little marks 27 cone wiggles, then 36 more. How many wiggles are marked?

27 + 36 = _____

Show your thinking.

2. Of 64 rice grains on the drum, 28 hop during a gentle tap. How many stay still?

64 − 28 = _____

Show your thinking.

3. Big records 38 drum taps on one page and 45 on another. How many taps are recorded?

38 + 45 = _____

Show your thinking.

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4. Little draws 72 waves and colors 46 of them. How many waves are left to color?

72 − 46 = _____

Show your thinking.

5. A speaker drawing shows 46 wire loops, then Little adds 29. How many loops are drawn?

46 + 29 = _____

Show your thinking.

6. A notebook has 91 cup-phone turns planned and 57 finished. How many turns remain?

91 − 57 = _____

Show your thinking.

WONDER DAILY

How Speakers Turn Waves Into Sound

A wiggle begins

Little heard a note from the speaker. A wiggly electric signal made something move. Could the notebook's smooth wave become a sound?

Push, then pull

A wire coil sat near a magnet. The changing signal made it push and pull. The attached paper cone moved back and forth.

A tiny hop

Little hummed near the rice bowl. Big gently tapped its side. The tight wrap vibrated, and rice hopped. A small motion became visible.

Two kinds of change

Rice hopped on the vibrating drumhead too. Faster wiggles make higher sounds. Bigger wiggles make louder sounds. Little kept the taps gentle.

Along a string

Big spoke into a paper cup. Its bottom vibrated. A taut string carried the motion to Little's cup, which moved the air nearby.

A wiggle reaches you

The cone sent pressure waves through air. Little's eardrum vibrated when they arrived. Where would the wiggle travel next?

Make a little book

1 Fold it into 8 boxes. Open it flat.

2 Fold short end to short end. Cut the thick line. Open it.

3 Fold the long way, pictures out. Push the ends in.

4 Fold into a book. Page 1 goes on the front!

Put a wiggle into words

How Speakers Turn Waves Into Sound · Curiosity day

Name _____ Date _____

Give a clear command for watching rice move.



Rice gives a clue



A drum can wiggle too



Another path for a wiggle

Draw the bowl before and after a gentle tap. Show the rice.

speaker · cone · coil · magnet · air · wave · ear · rice · gently

Hum ... tell where to make a soft sound

Watch ... tell which surface to observe

Draw ... tell how to show the motion

Does your command begin with an action word, such as Hum, Watch, or Draw?

How Speakers Turn Waves Into Sound

01 How can a wave become a sound?



A speaker gives a smooth signal a moving cone.

Look closer. Find the smooth wiggle beside the speaker.

A little explanation. Sound begins with vibration, a back-and-forth motion. A speaker turns a changing electric signal into motion. A pure note can follow a sine wave's smooth wiggle. Music uses many wiggles together.

Try this. Draw a smooth wave, then move your hand gently back and forth to follow its rhythm.

02 What makes the cone wiggle?



The coil moves the cone, and the cone moves air.

Look closer. Look for the wide paper cone on the speaker.

A little explanation. A speaker has a cone made of paper or plastic. A wire coil is attached behind it, near a magnet. The changing electric signal makes the coil push and pull against the magnet. The cone moves with it.

Try this. Use the diorama to follow the coil, the cone, and the air in that order.

03 Can you see a vibration?



A vibrating surface can make rice hop.

Look closer. Look just above the tight wrap.

A little explanation. Rice can hop when the surface under it vibrates. Try a toy drum, or a bowl with plastic wrap pulled tight across its top. A tiny tap makes the motion easier to spot.

Try this. Place a pinch of rice on the tight wrap or a toy drum. Hum or sing softly nearby and look closely. If the rice stays still, gently tap the bowl's side or the drum rim. Draw what changes. Keep Big beside you, sounds gentle, and rice on the table, away from mouths and ears. Leave devices closed; use no wall power or soldering, and let Big hold any magnet.

04 How does the wiggle reach your ear?



Pressure waves make an eardrum vibrate.

Look closer. Follow the air-wave marks toward the ear.

A little explanation. The cone squeezes air, then lets it spread out. These pressure changes travel as a wave. At your ear, the eardrum vibrates. Faster wiggles make a higher pitch. Bigger wiggles make a louder sound.

Try this. Draw a path from speaker to air to ear. Add one question about where the wiggle goes next.